

# pharo-agentic-browser Scripting API

**Multi-Agent Orchestration, Driven by Smalltalk**

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<https://github.com/mumez/pharo-agentic-browser>

**What is the Scripting API?**

An optional package that lets you **orchestrate AgenticBrowser topics from code** — no UI interaction required.

- Write a small Smalltalk script; AgenticBrowser creates topics, runs the agents, collects results between them, and blocks until everything completes
- A third interface alongside the Spec UI and Web UI

- **Routine AI workflows** — run the same multi-step workflow daily, or wire it into CI
- **Headless environments** — build and run topics with no display at all
- **Complex multi-agent coordination** — concurrent agents with result routing that's awkward to set up by hand in a UI
- **AI-authored orchestration** — the DSL is simple enough for a coding agent to write and run it itself via `st-eval`

# Features

- Just a few builder messages: `seq:`, `para:`, `topicBy:`, `agentBy:`
- Easy enough for a human **or an AI agent** to write and run with a single `eval`
- Results flow automatically between steps — no manual wiring

```
AgenticBrowser runBy: [ :builder |  
  builder seq: {  
    builder topicBy: [ :t | t prompt: 'List 3 Pharo Smalltalk features.' ].  
    builder topicBy: [ :t | t prompt: 'Write a one-sentence summary of previous features.' ].  
  } agentBy: [ :a | a claude ] ].
```

- Coordinate multiple **whole orchestrations**, not just topics within one
- Nest `seq:` / `para:` of orchestrations
- Nest groups inside groups
- Enables patterns like **Arena** (same task, different agents, pick the best) or fan-out research merged into one synthesis step

- Orchestrations (and groups) can be **saved and loaded** via Pharo's Fuel serializer
- Save before running to reuse a built configuration later
- Save after running to keep recorded results, then inspect or **resume** from a loaded copy



# Installation

Load the Scripting package alongside the core package:

```
Metacello new  
  baseline: 'AgenticBrowser';  
  repository: 'github://mumez/pharo-agentic-browser:main/src';  
  load: 'Scripting'.
```

Optionally, also load the test suite:

```
Metacello new  
  baseline: 'AgenticBrowser';  
  repository: 'github://mumez/pharo-agentic-browser:main/src';  
  load: 'Scripting-Tests'.
```

# Script Examples

`runBy:` builds and immediately runs the orchestration, blocking until done:

```
AgenticBrowser runBy: [ :builder |  
    builder seq: {  
        builder topicBy: [ :t |  
            t title: 'List Pharo features'.  
            t prompt: 'List 3 Pharo Smalltalk features in one sentence each.' ]  
        } agentBy: [ :a | a claude ] ].
```

`scriptBy:` builds without running, for later inspection or a `forkRun`.

Each topic's result flows into the next topic's prompt:

```
AgenticBrowser runBy: [ :builder |  
  builder seq: {  
    builder topicBy: [ :t |  
      t prompt: 'List 3 Pharo Smalltalk features in one sentence each.' ].  
    builder topicBy: [ :t |  
      t prompt: 'Summarize the feature list from the previous step in one sentence.' ]  
  } agentBy: [ :a | a claude ] ].
```

Topics run concurrently; results are combined for the next step:

```
AgenticBrowser runBy: [ :builder |  
  builder para: {  
    builder topicBy: [ :t | t prompt: 'List 3 Pharo Smalltalk language features.' ].  
    builder topicBy: [ :t | t prompt: 'List 3 Pharo Smalltalk development tools.' ]  
  } agentBy: [ :a | a claude ] ].
```

**seq:** and **para:** steps can be mixed freely — e.g. parallel research → sequential synthesis.

Run two candidates with different agents, then pick the winner:

```
AgenticBrowser groupRunBy: [ :groupBuilder |
  groupBuilder para: {
    groupBuilder orchestrationBy: [ :b |
      b seq: { b topicBy: [ :t | t prompt: 'Implement feature XXX.'. t goal: 'all tests pass' ] }
      agentBy: [ :a | a claude ] ].
    groupBuilder orchestrationBy: [ :b |
      b seq: { b topicBy: [ :t | t prompt: 'Implement feature XXX.'. t goal: 'all tests pass' ] }
      agentBy: [ :a | a codex ] ]
  }.
  groupBuilder singleTopicBy: [ :t | t prompt: 'Pick the best implementation above and open a PR.' ]
  agentBy: [ :a | a kilo ]
].
```

A full worked example is available in the repo docs:

[to-do-list-orchestration-script.md](#)

- Builds a complete **Spec2 To-do list app** in Pharo from scratch, with TDD
- **6 agents, 7 phases:** setup → parallel research → design → implementation → UI testing → review → documentation
- Mixes `seq:` / `para:`, lightweight vs. full models per phase, and a `goal:` to drive the implementation step until all tests pass
- A copy-paste-ready script — shows how far a single orchestration script can go



# **AI-Authored Orchestration**

Agent skill that lets the **agent itself write the orchestration script** — you describe the feature, it builds the DSL.

- Ask for a feature in plain language; the skill turns it into a runnable Scripting DSL orchestration targeting your own project
- Composes a sensible phase pipeline automatically: **plan (if needed)** → **TDD implementation** → **test** → **lint & style review**
- Previews the generated script as `docs/scripting-features/feature-<name>.scripting.md` before anything touches your codebase

- Nothing executes until you **explicitly approve** the previewed script
- Once approved, it's run via `st-eval` against the target repo ( `sharedDirectoryPath:` )
- While it runs, watch progress live in the **Spec UI** or **Web UI** — same as any orchestration

If a step stalls or times out, the skill retries automatically. You can also inspect and retry manually through `AbOrchestrationManager` .

The skill isn't just a demo — it has shipped real features.

- [RediStick](#)'s Time Series support was implemented by **running the orchestration script this skill generated**
- Generated script example: [scripting-features/feature-ts-mrange-mrevrange.scripting.md](#)
- Same shape as the To-Do app example — plan/implement/test/review — applied to a real production codebase

# Execution Variations

**Synchronous** — `runBy:` / `script run` blocks until everything completes.

**Background** — for long-running orchestrations:

```
script forkRunThen: [ :orc | Transcript crShow: 'Done: ', orc result ].  
"... later, if needed:"  
script isRunning.
```

`forkRun` is a shorthand for `forkRunThen: [ :orc | ]`.

While a background orchestration runs, open the **Spec UI** or **Web UI** to watch it live — think messages, permission confirmations, and topic progress all show up there.

**Cancel** a running orchestration:

```
script terminate.
```

**Resume** after a step times out (continues from the first incomplete step, reusing recorded results):

```
script resume.
```

**Save & Load**



Persist an orchestration (or group) via Fuel — before or after running:

```
script save.                                "-> <sharedDirectoryPath>/<name>.fuel"
script saveTo: '/path/to/my-script.fuel' asFileReference.

loaded := AbTopicOrchestration loadFrom: '/path/to/my-script.fuel'.
loaded result.                               "results from the saved run"
loaded resume.                              "continue if the saved run stopped early"
```

A saved orchestration can also be loaded straight into a new group with

```
orchestrationLoadFrom: .
```

# Settings

| Setting                                               | Default | Scope                                        |
|-------------------------------------------------------|---------|----------------------------------------------|
| <code>orchestrationStepWaitTimeoutSeconds</code>      | 900 s   | Per topic step                               |
| <code>orchestrationGroupItemWaitTimeoutSeconds</code> | 3600 s  | Per group item running in <code>para:</code> |

Adjust globally or per orchestration/group:

```
script settings orchestrationStepWaitTimeoutSeconds: 1800.
```

| Setting                                        | Default            | Effect                                                                    |
|------------------------------------------------|--------------------|---------------------------------------------------------------------------|
| <code>lingerOrchestrationTopicsAfterRun</code> | <code>false</code> | Whether topics stay in the Topic Manager after the orchestration finishes |

- `false` — topics are removed from the Topic Manager once the run completes
- `true` — topics remain, so you can review the whole run's progress from the UI afterward

```
script settings lingerOrchestrationTopicsAfterRun: true.
```

Set it to `true` when you want to go back through the Spec UI or Web UI later and retrace how the result was reached.

# Summary

The **Scripting API** brings headless, code-driven orchestration to AgenticBrowser:

- **Simple DSL** — `seq:` / `para:` / `topicBy:` / `agentBy:`, easy for humans and AI agents alike
- **Orchestration Groups** — compose whole workflows in parallel, in sequence, or nested
- **Result routing** — automatic, no manual wiring between steps
- **Flexible execution** — synchronous or background, with cancel and resume
- **Fuel persistence** — save and reload orchestrations and their results

# Feedback and contributions are welcome!

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